



LATERAL MOVEMENT EXERCISE

Technical Report & Documentation

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Program: Red Teaming Internship

Week: Week 3

Tools Used:- Impacket (psexec.py), Covenant / Alternative C2



1. Objective

The objective of this task is to simulate lateral movement within a compromised network environment. Using Impacket's psexec.py tool, the intern will pivot between hosts, establish persistence via a scheduled task backdoor, and document each step in alignment with MITRE ATT&CK framework techniques T1021 and T1053.

2. Lab Environment Setup

The following lab environment was configured prior to conducting the exercise. All activities were performed in an isolated virtual network with no connection to production systems.

2.1 Environment Overview

Screenshots below show the lab setup including the virtual machines, network configuration, and tools installed before the exercise began.

2.2 Tools Installed

Tool	Version / Source	Purpose
Impacket	github.com/fortra/impacket	Lateral movement via psexec.py
Covenant / Alternative C2	github.com/cobbr/Covenant	Command & Control framework
Kali Linux	Latest	Attacker machine OS
Windows Server / VM	Target OS	Pivot target host

3. Pivoting via Impacket psexec.py

Lateral movement was performed using Impacket's psexec.py script, which leverages SMB to execute commands on a remote Windows host using compromised credentials. This technique maps to MITRE ATT&CK T1021.002 – Remote Services: SMB/Windows Admin Shares.

3.1 Credential Acquisition

Prior to lateral movement, valid credentials were obtained from the initially compromised host. These were used to authenticate to the target host.



3.2 Executing psexec.py

The following command was used to pivot from the attacker machine to the target host:

```
python3 psexec.py <domain>/<username>:<password>@<target_IP>
```

3.3 Lateral Movement Path Summary

The pivot path from the attacker machine to the final target is summarised below (50 words):

Attacker machine (Kali Linux) gained initial foothold on Host A via exploitation. Harvested credentials were used with Impacket psexec.py to authenticate to Host B over SMB, establishing a remote SYSTEM shell. This allowed full control of the target, completing the lateral movement chain from Host A to Host B.

4. Persistence via Scheduled Task

To maintain persistent access on the compromised target, a scheduled task was created to execute a malicious payload at a defined interval. This technique maps to MITRE ATT&CK T1053.005 – Scheduled Task/Job: Scheduled Task.

4.1 Creating the Scheduled Task

The following command was executed on the target host to register a scheduled task that runs the payload daily:

```
schtasks /create /tn "SystemUpdate" /tr "C:\payload.exe" /sc daily /st 09:00
```

4.2 Persistence Log

The table below documents the persistence technique applied during this exercise:

Technique	Tactic	MITRE ID	Description	Notes
Scheduled Task	Persistence	T1053.005	Creates scheduled task to run payload on target host	Runs payload daily at 09:00
Registry Run Key (alt.)	Persistence	T1547.001	Alternative persistence via registry autostart	Not used in this task



5. Detection and Blue Team Notes

The following artefacts and indicators of compromise (IOCs) were generated during this exercise. A blue team analyst should look for these signs in SIEM or EDR logs.

Artefact / Log Source	Description	MITRE Reference
Windows Security Event 4624	Successful logon from attacker IP to target using harvested credentials	T1021
Windows Security Event 4698	Scheduled task creation logged in Event Viewer	T1053
SMB Traffic	Unusual SMB authentication between internal hosts at odd hours	T1021.002
psexec Artefacts	PSEXESVC service creation on target host	T1021.002

6. Findings and Recommendations

6.1 Key Findings

1. Lateral movement via psexec.py was successfully achieved using compromised credentials over SMB.
2. Persistence was established via a daily scheduled task that survives reboots.
3. Windows event logs (4624, 4698) provided clear artefacts for detection.
4. No network segmentation was in place between compromised and target hosts.

6.2 Recommendations

- Enforce network segmentation to limit SMB traffic between internal hosts.
- Enable and monitor Windows Event IDs 4624, 4648, 4698, and 7045 in a SIEM.
- Implement least-privilege access to prevent credential reuse across hosts.
- Deploy EDR solutions that alert on psexec/PSEXESVC service creation.
- Audit and whitelist scheduled tasks on all critical hosts.